Differential Equations — Reference

notation

Symbol	Meaning
y', $\dfrac{dy}{dx}$	first derivative of y
y'', $\dfrac{d^2y}{dx^2}$	second derivative
y''', $y^{(4)}$, $y^{(n)}$	third, fourth, n-th derivative
x or t	**independent variable** (the input)
y	**dependent variable** (the unknown function)
$y(x_0) = k_0$	**initial condition**: the value at x_0
c, c_1, c_2	arbitrary constants

If the independent variable is time, the book writes t, and y' means $\frac{dy}{dt}$.

terminology

Term	Meaning
differential equation (DE)	an equation with one or more derivatives of an unknown function
ordinary (ODE)	the unknown is a function of one variable
partial (PDE)	the unknown has two or more variables (partial derivatives)
order	the highest derivative in the equation
linear	y and its derivatives appear to the first power only, not multiplied together, not inside other functions
autonomous	$y' = f(y)$: no x on the right side
solution	a function that makes the equation true on an open interval
general solution	the family of all solutions, with constants c
particular solution	one member of the family, with numbers for c
explicit solution	$y =$ a formula in x
implicit solution	an equation in x and y, not solved for y
initial value problem (IVP)	a DE plus initial conditions
interval of validity	the largest open interval that contains x_0 on which the solution works
solution curve	the graph of a solution
integral curve	a curve made of solution curves (a circle, for example)
direction field	short slope marks drawn on a grid
isocline	a curve where the slope $f(x,y)$ is one constant
equilibrium solution	a constant solution $y = k$

linear or not? — the checklist

A linear equation can be written as

$$a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

Allowed (still linear)	Not allowed (nonlinear)
coefficients in x: x^2y'', e^xy'	a power of y: y^2, $(y')^3$, $\sqrt{y}$
any $g(x)$ on the right: $\sin x$	a product: yy', yy''
a constant coefficient: $3y'$	y inside a function: $\sin y$, e^y

calculus tables

Derivatives (a, n constants):

Function	Derivative	Note
x^n	nx^{n-1}	power rule
e^{ax}	ae^{ax}	chain rule
$\ln x$	$\dfrac{1}{x}$	
$\sin(ax)$	$a\cos(ax)$	
$\cos(ax)$	$-a\sin(ax)$	
$\tan x$	$\sec^2 x$	
fg	$f'g + fg'$	product rule
$\dfrac{f}{g}$	$\dfrac{f'g - fg'}{g^2}$	quotient rule
$f(g(x))$	$f'(g(x))\,g'(x)$	chain rule

Implicit differentiation: differentiate both sides in x. Each y term gets a factor y': $\frac{d}{dx}(y^2) = 2yy'$.

Integrals (add $+c$):

Function	Antiderivative
x^n, $n \neq -1$	$\dfrac{x^{n+1}}{n+1}$
$\dfrac{1}{x}$	$\ln\lvert x\rvert$
e^{ax}	$\dfrac{e^{ax}}{a}$
$\sin(ax)$	$-\dfrac{\cos(ax)}{a}$
$\cos(ax)$	$\dfrac{\sin(ax)}{a}$
$\sec^2 x$	$\tan x$
$\dfrac{1}{1+x^2}$	$\arctan x$

Integration by parts: $\int u\,dv = uv - \int v\,du$. Pick u = the part that gets simpler when you differentiate (x, x^2, $\ln x$).
u-substitution: $\int f\big(g(x)\big)g'(x)\,dx = \int f(u)\,du$ with $u = g(x)$.

algebra tools

quadratic formula	$r = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
two equations, two constants	solve one for c_1, substitute into the other
$e^{(\cdot)}$ is never 0	divide it out of an equation
$\ln$ undoes e	$e^a = b \Rightarrow a = \ln b$

formula sheet

1–3. Classification

order	highest derivative
linear form	$a_n(x)y^{(n)} + \cdots + a_0(x)y = g(x)$
normal form, first order	$y' = f(x,y)$
autonomous, first order	$y' = f(y)$

4–5. Solutions

verify	substitute y, y', …; both sides must match
try $y = e^{rx}$	$y' = re^{rx}$, $y'' = r^2e^{rx}$
try $y = x^r$	$y' = rx^{r-1}$, $y'' = r(r-1)x^{r-2}$

6–8. General solutions and IVPs

$y' = f(x)$	$y = \int f(x)\,dx + c$
$y^{(n)} = f(x)$	integrate n times, n constants
order n IVP	needs n conditions: $y(x_0), \ldots, y^{(n-1)}(x_0)$

9. Models

growth or decay (Malthus)	$P' = aP$, $P = P_0e^{at}$
logistic (Verhulst)	$P' = aP(1 - \alpha P)$, $P \to \frac{1}{\alpha}$
Newton's law of cooling	$T' = -k(T - T_m)$
cooling solution	$T = T_m + (T_0 - T_m)e^{-kt}$
glucose with injection	$G' = -\lambda G + r$
epidemic	$I' = rI(S - I)$
free fall	$y'' = -g$
free fall solution	$y = -\dfrac{g}{2}t^2 + v_0t + y_0$

10–12. Direction fields

slope at (x_0, y_0)	$m = f(x_0, y_0)$
isocline of slope c	$f(x,y) = c$
equilibrium of $y' = f(y)$	$f(k) = 0 \Rightarrow y = k$
stable equilibrium	arrows on both sides point toward it
unstable equilibrium	arrows on both sides point away
semistable equilibrium	toward on one side, away on the other

which tool? — given → find

The number in parentheses is the concept section.

Given → find	Do this	
an equation → ODE or PDE	count the independent variables	(1)
an equation → order	highest derivative	(2)
an equation → linear or not	the checklist	(3)
a function and a DE → is it a solution?	differentiate, substitute	(4)
a DE → values of r	try e^{rx} or x^r	(5)
DEs and functions → match	count constants, then verify	(5)
$y^{(n)} = f(x)$ → general solution	integrate n times	(6)
general solution, conditions → c	substitute, solve	(7)
a solution → interval of validity	remove bad x, keep x_0's piece	(8)
words → a DE	phrase table	(9)
$y' = f(x,y)$ → slope at a point	evaluate f	(10)
a field → its equation	match the features	(10)
slope c → where on the field	isocline $f(x,y) = c$	(11)
$y' = f(y)$ → constant solutions	solve $f(y) = 0$	(12)
$y(0)$ → long-term behavior	phase line	(12)

Differential Equations — Concepts

1. what a differential equation is

A **differential equation** (DE) is an equation that contains one or more derivatives of an unknown function. A solution is a *function*, not a number. Derivatives are rates of change. So a DE is a statement about how a quantity changes. Example: $P' = 0.02P$ says "the population grows at 2% of its size per unit time."

- **Ordinary** DE (ODE): the unknown is a function of one variable, $y(x)$. Derivatives are y', y''.
- **Partial** DE (PDE): the unknown has two or more variables, $u(x,t)$. Derivatives are $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial t}$.

Dependent variable: the unknown function (the one that is differentiated).
Independent variable: what it is differentiated with respect to.

Worked: name the type and the variables

1. $\frac{dy}{dt} = 3y - t$
One variable t ⇒ ODE. Dependent: y. Independent: t.

↓

2. $\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2}$
Two variables x, t ⇒ PDE. Dependent: u. Independent: x and t.

↓

3. $x^2\frac{d^2w}{dx^2} + w = 0$
One variable x ⇒ ODE. Dependent: w *(it is the one that is differentiated)*. Independent: x.

2. order

The **order** is the order of the highest derivative in the equation.
A power is not an order. $(y')^3$ is a first derivative cubed: order 1. y''' is a third derivative: order 3.
The highest derivative wins, even if a lower one has a bigger power. $y'' + (y')^5 = 0$ is order 2.

Worked: find the order of each equation

1. $y' + 3y = e^x$
Highest derivative: y' ⇒ order 1

↓

2. $(y')^4 + y = 0$
Only y' appears (raised to a power) ⇒ order 1

↓

3. $y'' - x(y')^3 = \sin x$
Highest derivative: y'' ⇒ order 2

↓

4. $\frac{d^3y}{dx^3} + \left(\frac{d^2y}{dx^2}\right)^2 = x$
Highest derivative: $\frac{d^3y}{dx^3}$ ⇒ order 3

3. linear or nonlinear

A DE is **linear** if it can be written as

$$a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

The coefficients $a_k(x)$ and the right side $g(x)$ can be any functions of x. Only y and its derivatives must be "plain".
Test: look at each term that contains y or a derivative of y. If any such term has a power, a product of two y terms, or y inside a function, the DE is **nonlinear**.
The x parts never decide. x^2y'', e^xy', $\sin(x)\,y$ are all linear terms.

Worked: linear or nonlinear?

1. $x^2y'' + xy' + y = \cos x$
y'', y', y each to the first power. The coefficients are in x only. linear

↓

2. $y' + y^2 = x$
y^2 is a power of y. nonlinear

↓

3. $yy' = 3$
$y \cdot y'$ is a product of two y terms. nonlinear

↓

4. $y'' + \sin y = 0$
y is inside sin. nonlinear

↓

5. $e^xy''' - 4y = x^3$
e^x is a coefficient in x. y''' and y are plain. linear

↓

6. $\frac{dy}{dx} = x + y$
Move y to the left: $y' - y = x$. Plain terms. linear

4. solutions, and how to verify one

A **solution** of a DE is a function that makes the equation true for every x in an open interval. It must have the derivatives that the DE uses.
Verify procedure:

1. Find each derivative the DE needs (y', y'', ...).
2. Substitute y and the derivatives into the left side.
3. Simplify.
4. Compare with the right side. They must be equal for all x in an interval.

Equal at a few points only is not a solution. $y = x^2$ makes $y' = 2y$ true at $x = 0$ and $x = 1$ only.
Constants c stay as letters. If the function has c, the check must work for every value of c.

Which way? — verify or find

Given → find	Do this
a function and a DE → yes or no	Differentiate the function and substitute.
a DE only → the function	Solve: integrate (section 6), or later methods.

Worked: verify $y = \frac{x^2}{3} + \frac{1}{x}$ solves $xy' + y = x^2$

1. DERIVATIVE
$y' = \frac{2x}{3} - \frac{1}{x^2}$ *($\frac{1}{x} = x^{-1}$, derivative $-x^{-2}$)*

↓

2. THE TERM xy'
$x\left(\frac{2x}{3} - \frac{1}{x^2}\right) = \frac{2x^2}{3} - \frac{1}{x}$

↓

3. ADD y
$xy' + y = \frac{2x^2}{3} - \frac{1}{x} + \frac{x^2}{3} + \frac{1}{x}$

↓

4. COMBINE LIKE TERMS
x^2 terms: $\frac{2}{3} + \frac{1}{3} = 1$. $\frac{1}{x}$ terms: $-\frac{1}{x} + \frac{1}{x} = 0$. Result: x^2

↓

5. COMPARE
Left $= x^2 =$ right ✓ y is undefined at $x = 0$, so it is a solution on $(0, \infty)$ and on $(-\infty, 0)$.

Worked: verify $y = ce^{2x}$ solves $y' = 2y$ for every c

1. DERIVATIVE
$y' = c \cdot 2e^{2x} = 2ce^{2x}$

↓

2. RIGHT SIDE
$2y = 2 \cdot ce^{2x} = 2ce^{2x}$

↓

3. COMPARE
$2ce^{2x} = 2ce^{2x}$ for all x and all c ✓ Interval: $(-\infty, \infty)$

Worked: (second order) verify $y = c_1e^x + c_2e^{-2x}$ solves $y'' + y' - 2y = 0$

1. FIRST DERIVATIVE
$y' = c_1e^x - 2c_2e^{-2x}$

↓

2. SECOND DERIVATIVE
$y'' = c_1e^x + 4c_2e^{-2x}$ *($-2 \cdot -2 = 4$)*

↓

3. SUBSTITUTE
$\left(c_1e^x + 4c_2e^{-2x}\right) + \left(c_1e^x - 2c_2e^{-2x}\right) - 2\left(c_1e^x + c_2e^{-2x}\right)$

↓

4. COLLECT THE e^x TERMS
$c_1 + c_1 - 2c_1 = 0$

↓

5. COLLECT THE e^{-2x} TERMS
$4c_2 - 2c_2 - 2c_2 = 0$

↓

6. COMPARE
Left $= 0 =$ right ✓

Worked: (a failure) is $y = x^2$ a solution of $y' = 2y$?

1. DERIVATIVE
$y' = 2x$

↓

2. RIGHT SIDE
$2y = 2x^2$

↓

3. COMPARE
$2x = 2x^2$ only at $x = 0$ and $x = 1$. Not on an interval ⇒ not a solution

5. finding r, and matching equations to solutions

Some problems give a form, like $y = e^{rx}$ or $y = x^r$, and ask for the values of r that make it a solution.

Procedure:

1. Differentiate the guess, with r as a letter.
2. Substitute into the DE.
3. Factor out the common function (e^{rx} or x^r).
4. That function is not zero, so the other factor must be 0. Solve for r.

Worked: find r so that $y = e^{rx}$ solves $y'' - y' - 6y = 0$

1. DERIVATIVES
$y' = re^{rx}, \quad y'' = r^2e^{rx}$

↓

2. SUBSTITUTE
$r^2e^{rx} - re^{rx} - 6e^{rx} = 0$

↓

3. FACTOR OUT e^{rx}
$e^{rx}\left(r^2 - r - 6\right) = 0$

↓

4. DIVIDE BY e^{rx}
$e^{rx} \neq 0$, so $r^2 - r - 6 = 0$

↓

5. FACTOR
$(r-3)(r+2) = 0 \Rightarrow r = 3$ or $r = -2$

↓

6. CHECK $r = 3$
$3^2 - 3 - 6 = 9 - 3 - 6 = 0$ ✓

Worked: find r so that $y = x^r$ solves $x^2y'' - 4xy' + 6y = 0$, $x > 0$

1. DERIVATIVES
$y' = rx^{r-1}, \quad y'' = r(r-1)x^{r-2}$

↓

2. THE TERM x^2y''
$x^2 \cdot r(r-1)x^{r-2} = r(r-1)x^r$ *(add the exponents: $2 + (r-2) = r$)*

↓

3. THE TERM xy'
$x \cdot rx^{r-1} = rx^r$

↓

4. SUBSTITUTE
$r(r-1)x^r - 4rx^r + 6x^r = 0$

↓

5. FACTOR OUT x^r
$x^r\left(r(r-1) - 4r + 6\right) = 0 \Rightarrow r^2 - r - 4r + 6 = 0$

↓

6. SIMPLIFY AND FACTOR
$r^2 - 5r + 6 = (r-2)(r-3) = 0 \Rightarrow r = 2$ or $r = 3$

Matching equations to solutions

A matching problem gives a list of DEs and a list of functions. Sort first with fast clues. Then verify each pair with the procedure from section 4.

1. **Count the constants.** A general solution of an order n DE has n constants. One $c \Rightarrow$ first order. c_1 and $c_2 \Rightarrow$ second order.
2. **Read the form of the function.** Use the clue table below to guess the partner.
3. **Verify the guess.** Differentiate and substitute. A guess is not an answer until it checks.
4. **Cross off the pair.** Each remaining choice is easier.

The function has	Look for the DE
ce^{kx} only	$y' = ky$
$k + ce^{-ax}$ (a constant plus an exponential)	$y' = a(k-y)$ *(equilibrium at k)*
$c_1e^{r_1x} + c_2e^{r_2x}$	$y'' + by' + cy = 0$ with $r^2 + br + c = 0$ at r_1, r_2
$c_1\cos(kx) + c_2\sin(kx)$	$y'' + k^2y = 0$
$c_1x^{r_1} + c_2x^{r_2}$	$x^2y'' + bxy' + cy = 0$ *(test with x^r)*
a polynomial plus c	$y' = f(x)$: integrate to check

Same order is not enough. Two second-order DEs can both have two-constant solutions. Always finish with the substitution check.

Worked: match (A) $y' = -3y$, (B) $y'' + 9y = 0$, (C) $y'' - y' - 2y = 0$, (D) $y' = 2(4 - y)$ to (1) $c_1 \cos 3x + c_2 \sin 3x$, (2) ce^{-3x}, (3) $4 + ce^{-2x}$, (4) $c_1 e^{2x} + c_2 e^{-x}$

1. COUNT THE CONSTANTS
(1) and (4) have two ⇒ they go with (B) or (C).
(2) and (3) have one ⇒ they go with (A) or (D).

↓

2. GUESS FROM THE FORM
(1) is cosine and sine of $3x$ ⇒ (B), since $k^2 = 9$. (2) is ce^{kx} with $k = -3$ ⇒ (A).

↓

3. VERIFY (1) IN (B): DERIVATIVES
$y' = -3c_1 \sin 3x + 3c_2 \cos 3x$. $y'' = -9c_1 \cos 3x - 9c_2 \sin 3x$

↓

4. VERIFY (1) IN (B): SUBSTITUTE
$y'' = -9(c_1 \cos 3x + c_2 \sin 3x) = -9y$. So $y'' + 9y = -9y + 9y = 0$ ✓

↓

5. VERIFY (2) IN (A)
$y' = -3ce^{-3x} = -3(ce^{-3x}) = -3y$ ✓

↓

6. VERIFY (4) IN (C), THE LAST TWO-CONSTANT PAIR
Test each exponent in $r^2 - r - 2$: $r = 2$: $4 - 2 - 2 = 0$ ✓ $r = -1$: $1 + 1 - 2 = 0$ ✓

↓

7. VERIFY (3) IN (D): LEFT SIDE
$y' = -2ce^{-2x}$ *(the 4 is constant, its derivative is 0)*

↓

8. VERIFY (3) IN (D): RIGHT SIDE
$2(4 - y) = 2(4 - 4 - ce^{-2x}) = 2(-ce^{-2x}) = -2ce^{-2x}$. Same ✓

↓

9. RESULT
A–2, B–1, C–4, D–3

Worked: (no constants) which of $y = e^{-x}$ or $y = xe^{-x}$ solves $y'' + 2y' + y = 0$?

1. COUNTING DOES NOT HELP
Neither has a constant. Verify both.

↓

2. $y = e^{-x}$
$y' = -e^{-x}$, $y'' = e^{-x}$. $e^{-x} - 2e^{-x} + e^{-x} = 0$ ✓

↓

3. $y = xe^{-x}$: DERIVATIVES
$y' = e^{-x} - xe^{-x}$ *(product rule)*. $y'' = -e^{-x} - e^{-x} + xe^{-x} = -2e^{-x} + xe^{-x}$

↓

4. $y = xe^{-x}$: SUBSTITUTE
$(-2e^{-x} + xe^{-x}) + 2(e^{-x} - xe^{-x}) + xe^{-x}$

↓

5. COLLECT
e^{-x}: $-2 + 2 = 0$. xe^{-x}: $1 - 2 + 1 = 0$. Sum $= 0$ ✓

↓

6. RESULT
Both are solutions. *(a matching list can have more than one answer per equation; check every pair)*

6. general solutions by direct integration

The simplest DE has only x on the right: $y' = f(x)$. Integrate both sides:

$$y = \int f(x)\,dx + c$$

For $y^{(n)} = f(x)$, integrate n times. Each integration adds a new constant. So the **general solution** of an order n equation has n arbitrary constants.
Add the constant at each integration, not only at the end. The constant from the first integration becomes $c_1 x$ after the second.

Worked: general solution of $y' = x^3$

1. INTEGRATE
$y = \int x^3\,dx = \frac{x^4}{4} + c$

↓

2. CHECK
$\frac{d}{dx}\left(\frac{x^4}{4} + c\right) = x^3$ ✓

Worked: general solution of $y'' = \cos(2x)$

1. FIRST INTEGRATION
$y' = \int \cos(2x)\,dx = \frac{\sin(2x)}{2} + c_1$

↓

2. SECOND INTEGRATION
$y = \int \left(\frac{\sin(2x)}{2} + c_1\right) dx$

↓

3. EACH TERM
$\frac{1}{2} \cdot \left(-\frac{\cos(2x)}{2}\right) = -\frac{\cos(2x)}{4}$. $\int c_1\,dx = c_1 x$.

↓

4. ASSEMBLE
$y = -\frac{\cos(2x)}{4} + c_1 x + c_2$ *(two constants for order 2)*

Worked: (by parts) general solution of $y' = xe^{3x}$

1. PICK u AND dv
$u = x$, $dv = e^{3x}\,dx$ *(x gets simpler when differentiated)*

↓

2. FIND du AND v
$du = dx$, $v = \frac{e^{3x}}{3}$

↓

3. PARTS FORMULA $uv - \int v\,du$
$\frac{xe^{3x}}{3} - \int \frac{e^{3x}}{3}\,dx$

↓

4. FINISH THE INTEGRAL
$\int \frac{e^{3x}}{3}\,dx = \frac{e^{3x}}{9}$

↓

5. ASSEMBLE
$y = \frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + c$

↓

6. CHECK
$y' = \frac{e^{3x}}{3} + xe^{3x} - \frac{e^{3x}}{3} = xe^{3x}$ ✓ *(product rule on the first term)*

7. initial value problems

An **initial value problem** (IVP) is a DE plus conditions at one point x_0. The conditions pick one **particular solution** out of the general family.
An order n IVP needs n conditions: $y(x_0), y'(x_0), \ldots, y^{(n-1)}(x_0)$.
Procedure:

1. Find the general solution (with c, or c_1, c_2).
2. For each condition, substitute x_0 and the given value. Each gives one equation.
3. Solve the equations for the constants.
4. Write the particular solution. Check each condition.

A y' condition goes into y', not into y. Differentiate the general solution first.

Which way? — general or particular

Given → find	Do this
general solution + conditions → particular	substitute x_0, solve for the constants
particular solution → its condition	evaluate it at x_0
a point the graph passes through	that is the condition: $(1,2)$ means $y(1) = 2$

Worked: $y' = 3x^2, \quad y(2) = 5$

1. GENERAL SOLUTION
$y = \int 3x^2\,dx = x^3 + c$

↓

2. SUBSTITUTE $x = 2, y = 5$
$5 = 2^3 + c = 8 + c$

↓

3. SOLVE
$c = 5 - 8 = -3$

↓

4. PARTICULAR SOLUTION
$y = x^3 - 3$. Check: $2^3 - 3 = 5$ ✓

Worked: $y'' = 6x, \quad y(0) = 1, \quad y'(0) = -2$

1. FIRST INTEGRATION
$y' = 3x^2 + c_1$

↓

2. USE $y'(0) = -2$
$-2 = 3(0)^2 + c_1 \Rightarrow c_1 = -2$. So $y' = 3x^2 - 2$

↓

3. SECOND INTEGRATION
$y = x^3 - 2x + c_2$

↓

4. USE $y(0) = 1$
$1 = 0 - 0 + c_2 \Rightarrow c_2 = 1$

↓

5. PARTICULAR SOLUTION
$y = x^3 - 2x + 1$

↓

6. CHECK
$y(0) = 1$ ✓ $\quad y'(0) = 0 - 2 = -2$ ✓ $\quad y'' = 6x$ ✓

Worked: general solution $y = c_1 e^x + c_2 e^{-2x}, \quad y(0) = 3, \quad y'(0) = 0$

1. DIFFERENTIATE THE GENERAL SOLUTION
$y' = c_1 e^x - 2c_2 e^{-2x}$

↓

2. CONDITION $y(0) = 3$
$e^0 = 1$: $\quad c_1 + c_2 = 3$

↓

3. CONDITION $y'(0) = 0$
$c_1 - 2c_2 = 0$

↓

4. SOLVE THE SECOND FOR c_1
$c_1 = 2c_2$

↓

5. SUBSTITUTE INTO THE FIRST
$2c_2 + c_2 = 3 \Rightarrow 3c_2 = 3 \Rightarrow c_2 = 1$

↓

6. BACK-SUBSTITUTE
$c_1 = 2(1) = 2$

↓

7. PARTICULAR SOLUTION
$y = 2e^x + e^{-2x}$

↓

8. CHECK
$y(0) = 2 + 1 = 3$ ✓ $\quad y'(0) = 2 - 2 = 0$ ✓

Worked: (first order family) $y = 3 + ce^{-2x}, \quad y(0) = 7$

1. SUBSTITUTE $x = 0, y = 7$
$7 = 3 + ce^0 = 3 + c$

↓

2. SOLVE
$c = 4 \Rightarrow y = 3 + 4e^{-2x}$

8. interval of validity; explicit and implicit

The **interval of validity** is the largest open interval that contains x_0 on which the solution is defined and solves the DE.

Procedure:

1. Find each x where the solution (or the DE) breaks: a zero denominator, a negative under a square root, ln of 0 or less.
2. Those x values cut the number line into intervals.
3. Keep the one interval that contains x_0.

Explicit solution: y is alone, $y = \ldots$. **Implicit solution**: an equation in x and y, like $x^2 + y^2 = 25$. Check an implicit solution with implicit differentiation.

Worked: $y = \frac{1}{x-2}$ ***and*** $y' = -y^2, \quad y(3) = 1$

1. VERIFY: DERIVATIVE
$y = (x-2)^{-1} \Rightarrow y' = -(x-2)^{-2} = -\dfrac{1}{(x-2)^2}$

↓

2. VERIFY: RIGHT SIDE
$-y^2 = -\dfrac{1}{(x-2)^2}$. Same ✓

↓

3. VERIFY: CONDITION
$y(3) = \dfrac{1}{3-2} = 1$ ✓

↓

4. WHERE IT BREAKS
$x - 2 = 0$ at $x = 2$. Intervals: $(-\infty, 2)$ and $(2, \infty)$

↓

5. KEEP THE ONE WITH $x_0 = 3$
Interval of validity: $(2, \infty)$ *(with* $y(0) = -\frac{1}{2}$ *instead, it is* $(-\infty, 2)$*)*

Worked: (implicit) $x^2 + y^2 = 25$ ***and*** $y' = -\frac{x}{y}$

1. DIFFERENTIATE BOTH SIDES IN x
$2x + 2yy' = 0$ *(chain rule:* $y^2 \to 2yy'$*)*

↓

2. SOLVE FOR y'
$2yy' = -2x \Rightarrow y' = -\dfrac{x}{y}$ ✓

↓

3. EXPLICIT PIECE THROUGH $(3, 4)$
$y > 0$ there, so $y = \sqrt{25 - x^2}$

↓

4. INTERVAL
Need $25 - x^2 > 0$ *(*$y = 0$ *makes* y' *undefined)* $\Rightarrow (-5, 5)$

9. models: from words to a DE

A **model** turns a description into an equation. "Rate of change" means a derivative.

Words	Math
the rate of change of y	$\dfrac{dy}{dt}$
is proportional to y	$= ky$
is proportional to the square of y	$= ky^2$
is inversely proportional to y	$= \dfrac{k}{y}$
is proportional to the difference of T and T_m	$= k(T - T_m)$
is proportional to the product of I and $S - I$	$= kI(S - I)$
decreases, decays, is absorbed	a minus sign, with $k > 0$
is added at a constant rate r	$+r$

Worked: write each model as a DE

1. A SUBSTANCE DECAYS AT A RATE PROPORTIONAL TO THE AMOUNT A
$\frac{dA}{dt} = -kA, \quad k > 0$ *("decays" gives the minus sign)*

↓

2. A RUMOR SPREADS AT A RATE PROPORTIONAL TO THE PRODUCT OF THOSE WHO HEARD IT, R, AND THOSE WHO DID NOT, IN A TOWN OF 5000
Did not hear: $5000 - R$. $\frac{dR}{dt} = kR(5000 - R)$

Worked: growth: $P' = 0.02P$, $P(0) = 1000$. Find $P(10)$

1. MATCH THE MODEL
$P' = aP$ with $a = 0.02 \Rightarrow P = P_0 e^{at}$

↓

2. PUT IN P_0
$P = 1000e^{0.02t}$

↓

3. EVALUATE AT $t = 10$
$0.02 \cdot 10 = 0.2$. $e^{0.2} = 1.2214$. $P(10) = 1000(1.2214) = 1221$

Worked: (reverse) a population doubles in 35 years. Find a

1. DOUBLING MEANS
$P(35) = 2P_0 \Rightarrow P_0 e^{35a} = 2P_0$

↓

2. DIVIDE BY P_0
$e^{35a} = 2$

↓

3. TAKE ln
$35a = \ln 2 = 0.6931$

↓

4. DIVIDE
$a = \frac{0.6931}{35} = 0.0198$ per year

Worked: cooling: $T' = -0.1(T - 20)$, $T(0) = 90$. Find $T(5)$

1. MATCH THE MODEL
$k = 0.1, \quad T_m = 20, \quad T_0 = 90$

↓

2. SOLUTION FORMULA
$T = T_m + (T_0 - T_m)e^{-kt} = 20 + 70e^{-0.1t}$

↓

3. EVALUATE AT $t = 5$
$e^{-0.5} = 0.6065$. $T(5) = 20 + 70(0.6065) = 20 + 42.46 = 62.5$

↓

4. LONG TERM
$e^{-0.1t} \to 0$, so $T \to 20$, the room temperature

Worked: free fall: $y'' = -9.8$, $y(0) = 24.5$ m, $y'(0) = 19.6$ m/s

1. FIRST INTEGRATION (VELOCITY)
$y' = -9.8t + c_1$. $y'(0) = 19.6 \Rightarrow c_1 = 19.6$

↓

2. SECOND INTEGRATION (HEIGHT)
$y = -4.9t^2 + 19.6t + c_2$. $y(0) = 24.5 \Rightarrow c_2 = 24.5$

↓

3. HIGHEST POINT: VELOCITY = 0
$-9.8t + 19.6 = 0 \Rightarrow t = 2$ s

↓

4. HEIGHT AT $t = 2$
$-4.9(4) + 19.6(2) + 24.5 = -19.6 + 39.2 + 24.5 = 44.1$ m

↓

5. GROUND: HEIGHT = 0
$-4.9t^2 + 19.6t + 24.5 = 0$

↓

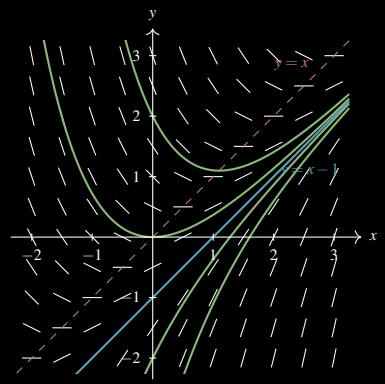
6. DIVIDE BY -4.9
$t^2 - 4t - 5 = 0$

↓

7. FACTOR
$(t - 5)(t + 1) = 0 \Rightarrow t = 5$ s *($t = -1$ is before the launch)*

10. direction fields

For a first order equation $y' = f(x, y)$, the right side gives the slope of the solution curve at each point. You can draw slopes without solving the DE.
Slope at a point: $m = f(x_0, y_0)$.
Direction field (slope field): at each point of a grid, draw a short segment with slope $f(x, y)$. Solution curves follow the segments, like water follows a current.



- The field above is for $y' = x - y$. Green: solution curves. **Blue**: the solution $y = x - 1$. Dashed red: the slope-0 line $y = x$.
- Every curve bends toward the blue line as x grows.

How to read a field and match it to an equation:

Feature of the field	What it tells you
slopes are the same along each horizontal row	f has no x: autonomous, $y' = f(y)$
slopes are the same along each vertical column	f has no y: $y' = f(x)$
flat segments along a line or curve	that is where $f(x, y) = 0$
a horizontal row of flat segments at $y = k$	equilibrium solution $y = k$
slope sign in each region	sign of f there

Which way? — equation or field

Given → find	Do this
equation → field	evaluate f at grid points, draw the segments
field → equation	test the features in the table above, one at a time
field and a start point → curve	start there, follow the segments both ways

Worked: slopes of $y' = x - y$ at five points

1. AT $(0,0)$
$m = 0 - 0 = 0$ (flat)

↓

2. AT $(1,0)$
$m = 1 - 0 = 1$ (up at $45°$)

↓

3. AT $(0,1)$
$m = 0 - 1 = -1$ (down at $45°$)

↓

4. AT $(2,-1)$
$m = 2 - (-1) = 3$ (steep up)

↓

5. AT $(-1,2)$
$m = -1 - 2 = -3$ (steep down)

↓

6. CHECK WITH THE PICTURE
Below the dashed line ($y < x$) the slopes are positive. Above it, negative ✓

Worked: match: which of $y' = y$, $y' = x$, $y' = x - y$ fits a field?

The field has: flat segments along the line $y = x$, slopes that change along each row and each column.

1. TEST $y' = y$
It is autonomous: each row would be the same slope. The field's rows change. no

↓

2. TEST $y' = x$
Each column would be the same slope, flat along $x = 0$. The field is flat along $y = x$. no

↓

3. TEST $y' = x - y$
Flat where $x - y = 0$, which is $y = x$. Both x and y change the slope. match

Worked: (reverse) sketch the solution of $y' = x - y$ through $(0,1)$

1. START POINT
At $(0,1)$ the slope is -1: the curve goes down to the right.

↓

2. FOLLOW THE SEGMENTS
It falls until it crosses $y = x$ (flat there), then rises.

↓

3. LONG TERM
It bends toward $y = x - 1$ as x grows. *(the exact solution is $x - 1 + 2e^{-x}$)*

11. isoclines

An **isocline** is a curve on which every segment has the same slope c:

$$f(x,y) = c$$

Draw a few isoclines first. Then put segments of slope c along each one. This is faster than point-by-point.

The isocline for $c = 0$ is where solution curves are flat: their peaks and valleys sit there.

Worked: isoclines of $y' = x - y$

1. SET $f = c$
$x - y = c \Rightarrow y = x - c$: lines of slope 1

↓

2. $c = 0$
$y = x$: flat segments

↓

3. $c = 1$
$y = x - 1$: segments of slope 1

↓

4. $c = -1$
$y = x + 1$: segments of slope -1

↓

5. NOTICE
On $y = x - 1$ the segments have slope 1, the same as the line. So the line is itself a solution (blue in section 10).

Worked: (reverse) where does $y' = x^2 + y^2$ have slope 4?

1. SET $f = 4$
$x^2 + y^2 = 4$

↓

2. READ THE CURVE
A circle of radius 2 about the origin

12. equilibrium solutions and long-term behavior

For an **autonomous** equation $y' = f(y)$, a constant $y = k$ is a solution if $f(k) = 0$. It is an **equilibrium solution**: a horizontal line of flat segments.

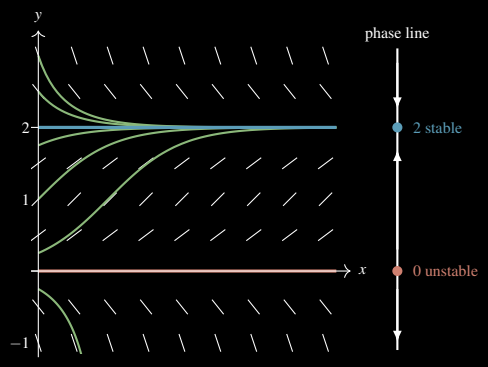
Phase line procedure:

1. Solve $f(y) = 0$. The roots are the equilibria.
2. Mark them on a vertical y line. They cut it into intervals.
3. Test one y in each interval: $f > 0$ means y increases (arrow up). $f < 0$ means it decreases (arrow down).
4. Classify each equilibrium by its arrows.

Type	Arrows
stable	both point toward it: nearby solutions go to it
unstable	both point away: nearby solutions leave it
semistable	toward on one side, away on the other

Long-term behavior: a solution starting at $y(0)$ follows the arrow of its interval. It goes to the next equilibrium in that direction, or to $\pm\infty$ if there is none.

Solution curves cannot cross an equilibrium line. A curve that starts between two equilibria stays between them.



The field above is for $y' = y(2 - y)$, with its phase line on the right.

Worked: equilibria, stability, and limits for $y' = y(2-y)$

1. SOLVE $f(y) = 0$
$y(2-y) = 0 \Rightarrow y = 0$ or $y = 2$

↓

2. TEST $y = -1$ (BELOW 0)
$f(-1) = (-1)(2-(-1)) = (-1)(3) = -3 < 0$: down

↓

3. TEST $y = 1$ (BETWEEN)
$f(1) = (1)(2-1) = 1 > 0$: up

↓

4. TEST $y = 3$ (ABOVE 2)
$f(3) = (3)(2-3) = -3 < 0$: down

↓

5. CLASSIFY $y = 2$
Up from below, down from above: both toward $\Rightarrow$ stable

↓

6. CLASSIFY $y = 0$
Down below, up above: both away $\Rightarrow$ unstable

↓

7. LIMITS AS $x \to \infty$
$y(0) = 1$ or $y(0) = 3$: $y \to 2$. $y(0) = 0$: stays 0. $y(0) = -1$: $y \to -\infty$.

Worked: (semistable) $y' = (y-1)^2$

1. SOLVE $f(y) = 0$
$(y-1)^2 = 0 \Rightarrow y = 1$ only

↓

2. TEST $y = 0$ (BELOW)
$f(0) = (0-1)^2 = 1 > 0$: up, toward 1

↓

3. TEST $y = 2$ (ABOVE)
$f(2) = (2-1)^2 = 1 > 0$: up, away from 1

↓

4. CLASSIFY
Toward from below, away above $\Rightarrow$ **semistable**

Worked: (reverse) build an autonomous DE with equilibria -1 (stable) and 3 (unstable)

1. ROOTS
Try $f(y) = a(y+1)(y-3)$

↓

2. TEST THE MIDDLE $y = 0$ WITH $a = 1$
$f(0) = (1)(-3) = -3 < 0$: down toward -1, away from 3 ✓

↓

3. TEST THE ENDS
$y = -2$: $f = (-1)(-5) = 5 > 0$, up toward -1 ✓ $y = 4$: $f = (5)(1) = 5 > 0$, away from 3 ✓

↓

4. RESULT
$y' = (y+1)(y-3)$

Worked: (no equilibrium) does $y' = x - y$ have a constant solution?

1. TRY $y = k$
Then $y' = 0$, so $0 = x - k$ for every x.

↓

2. CONCLUDE
That fails for all $x \neq k \Rightarrow$ no equilibrium. *(it is not autonomous; its straight-line solution $y = x - 1$ is not constant)*